

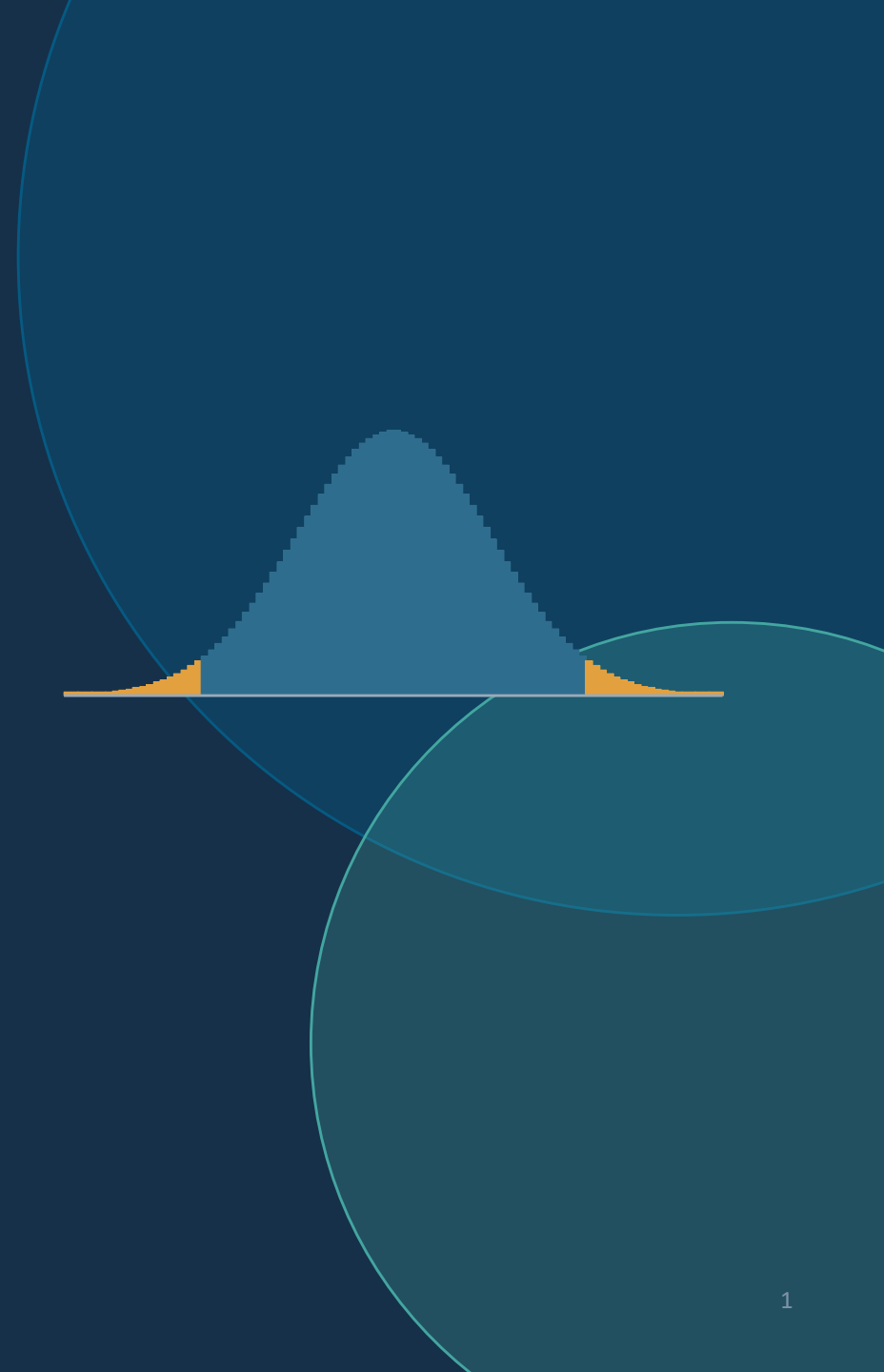
STATISTICS · PART 2

# Confidence Intervals & Hypothesis Testing

*From estimating where a parameter lives, to deciding yes or no.*

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A lecture deck for learners of inferential statistics



# Two halves of the same idea

Inference draws conclusions about a population from nothing but a sample. There are two ways to do it.

1

## Confidence Intervals

- A range where the parameter probably sits
- One population, then two populations
- Variance known vs unknown
- Margin of error and what controls it

2

## Hypothesis Testing

- A yes or no answer for decisions
- Null and alternative hypotheses
- Significance, rejection regions, p-values
- Type I and Type II errors

*Two full practical examples close each half: shoe inventory at Al Bundy's, and a gender pay gap study at Spark Fortress.*

PART 1

# Confidence Intervals

*Estimating where the truth lives.*

- Why a single number is never enough
- Known variance, unknown variance, one population, two
- Margin of error and the confidence tradeoff

## THE PROBLEM

# A point estimate is a lucky guess

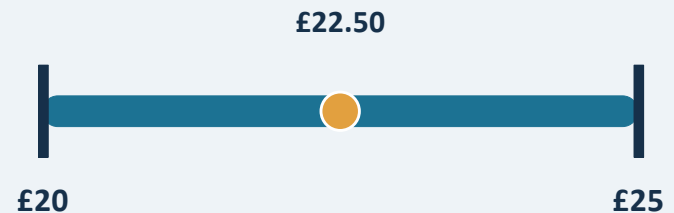
A survey covers 5% of the restaurants in London and reports an average meal price of 22.50 pounds.

- The estimate is probably close, but almost certainly not exact
- The true value sits somewhere around 22.50, not on it
- A single number hides the level of certainty behind it

**100% confidence is impossible without measuring the entire population.**

### A safer statement

**“The average meal in London costs between 20 and 25 pounds.”**



*A range is a more accurate picture of reality than a point.*

# How sure is sure?

The confidence level is written as  $1 - \alpha$ , where  $\alpha$  is a value between 0 and 1.

**90%**

$\alpha = 0.10$

Loosest of the three

**95%**

$\alpha = 0.05$

The accepted norm

**99%**

$\alpha = 0.01$

When mistakes are costly

## Reading it correctly

- 95% confident means: in 95% of cases, the true population parameter falls inside the interval
- There is still a 5% chance it sits outside, which happens when the sample deviates badly from the population

# One formula, three ingredients

$$\text{point estimate} \pm \text{reliability factor} \times \text{standard error}$$

1

## Point estimate

The value computed from the sample, such as  $\bar{x}$  for the mean. Sits at the centre of the interval.

2

## Reliability factor

z or t, pulled from a table. Determined by the confidence level.

3

## Standard error

How much sample means bounce around. Shrinks as the sample grows.

*Everything that follows is this formula with different pieces plugged in.*

# Is the population variance known?



## Variance KNOWN

- Use the z-statistic
- Pair it with the population standard deviation  $\sigma$
- Happens rarely in real work
- Needs a normal population, or a big enough sample



## Variance UNKNOWN

- Use the t-statistic
- Pair it with the sample standard deviation  $s$
- This is the normal state of affairs
- Follows Student's T distribution

Why statistics exists at all: population data is almost never available. Even Google, sitting on the largest dataset on earth, misses everyone on Bing, DuckDuckGo, Safari or incognito. Even that is still a sample.

# Confidence interval, variance known

$$\bar{X} \pm z_{\alpha/2} \times \sigma / \sqrt{n}$$

## Assumptions

- The population is normally distributed
- If it is not, a large sample lets the central limit theorem normalise the sample means
- The population standard deviation  $\sigma$  is genuinely known, usually from past data

## The problem

Estimate the average data scientist salary.

Sample size **n = 30**

Sample mean **\$100,200**

Population  $\sigma$  **\$15,000**

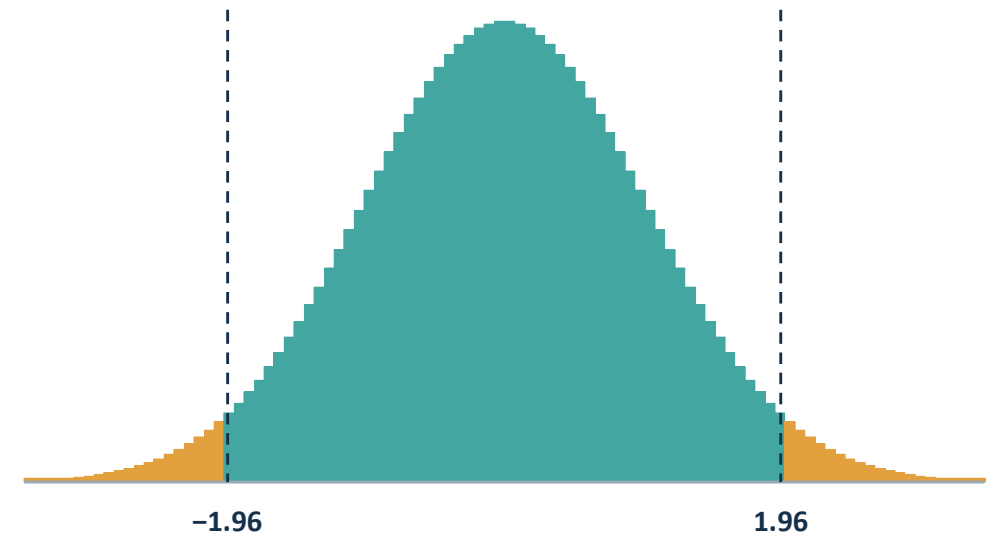
Standard error **\$2,739**



# Reading the standard normal table

- For 95% confidence,  $\alpha = 0.05$ , so  $\alpha/2 = 0.025$
- In the table, look for  $1 - 0.025 = 0.9750$
- The z is the row header plus the column header of that cell:  $1.9 + 0.06$
- That gives 1.96, commonly called the critical value

For 99% confidence,  $\alpha = 0.01$ . The value 0.995 is missing from the table, so round to the nearest available one:  $2.5 + 0.08 = 2.58$ .



95% of the area sits between the two cut lines. The shaded tails hold 2.5% each.

**1.96**

critical value at 95%

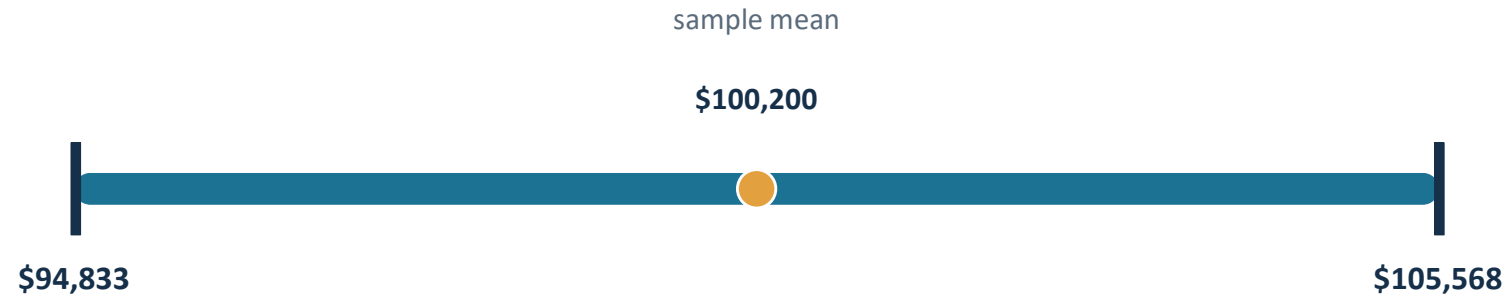
**2.58**

critical value at 99%

## WORKED RESULT

# The 95% interval for data scientist salary

$$100,200 \pm 1.96 \times 2,739$$

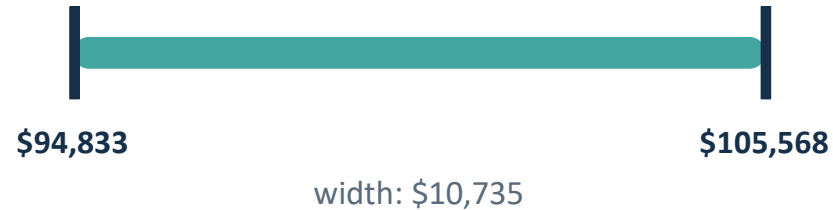


### Stating the result

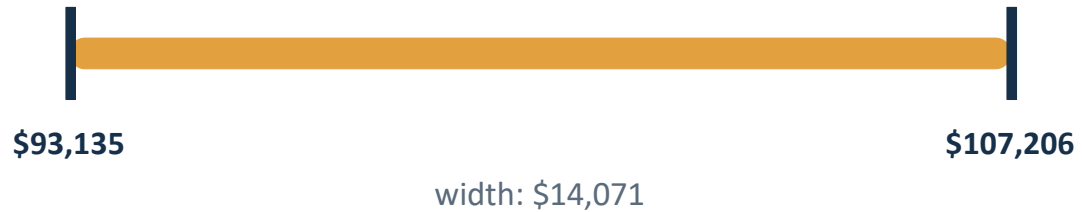
**“At 95% confidence, the average data scientist salary falls between \$94,833 and \$105,568.”**

# More confidence costs precision

## 95% confidence



## 99% confidence



### Why this happens

- A wider net catches the true mean more often, so higher confidence always widens the interval
- A narrow interval is a bolder claim, so it carries more uncertainty about being right
- Being more specific about the mean takes away from confidence in the statement

## INTUITION CHECK

# How wide is too wide?

Estimating the age of a student enrolled in an online statistics course, with no other information available:

**100%**

**0 to 118 years old**

Useless. Every living age has to be included to reach certainty.

**99%**

**10 to 70 years old**

Narrower, but still not insightful for this problem.

**95%**

**18 to 55 years old**

The accepted balance. Reasonable range, reasonable certainty.

**5%**

**exactly 25 years old**

Perfectly precise, but the confidence is too low to use.

*There is always a tradeoff, and it depends on the problem. 95% is the norm because it gives a usable range without giving up much accuracy.*

## A SHORT STORY

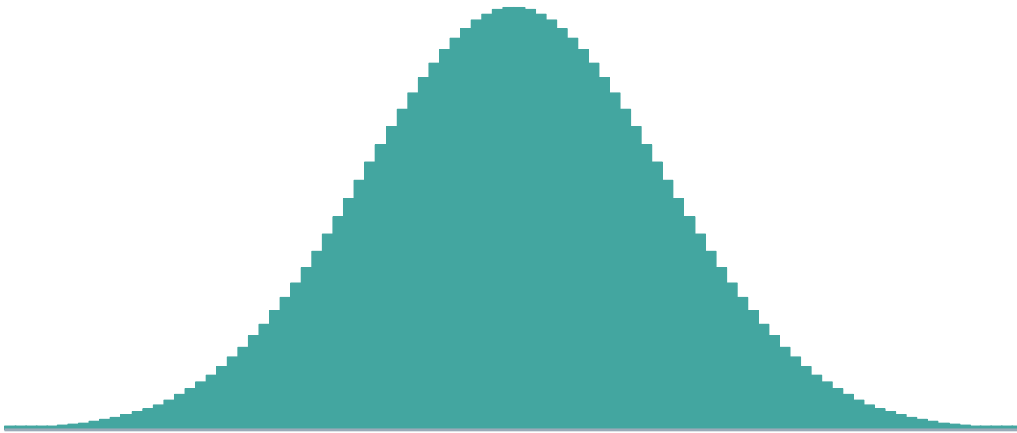
# The statistician who wasn't allowed a name

- William Gosset was an English statistician working for the Guinness brewery, developing methods to select the best yielding barley
- Big samples were tedious, so he built a way to draw small samples and still make meaningful predictions
- Guinness company policy barred him from signing his own papers, so he published everything under the pen name “Student”
- His friend Ronald Fisher built on the work and introduced the t-statistic. The name that stuck is Student's T

### Why it mattered

It opened up inference from small samples with unknown population variance, which describes most real statistical problems today.

# Same shape, fatter tails



Fatter tails allow higher dispersion, which is exactly what more uncertainty looks like.

**Rule of thumb: more than 50 observations, just use the z-table.**

## Key properties

- The t-statistic relates to the T distribution the same way z relates to the standard normal
- $t = (\bar{x} - \mu) / \text{standard error}$ , which is the same shape as the z formula
- Degrees of freedom are  $n - 1$ , so a sample of 20 has 19 df
- In the t-table, rows are degrees of freedom and columns are common alphas
- After 30 df, the values barely move and the table looks almost like the z-table

# Confidence interval, variance unknown

Variance known

$$\bar{X} \pm z_{\alpha/2} \times \sigma / \sqrt{n}$$

Variance unknown

$$\bar{X} \pm t_{n-1, \alpha/2} \times s / \sqrt{n}$$

Only two things change

1

**z becomes t**

The distribution is now Student's T, not the standard normal.

2

**$\sigma$  becomes s**

The population standard deviation is replaced by the sample standard deviation.

The pairing to memorise: variance known → population  $\sigma$  goes with z. Variance unknown → sample s goes with t.

## WORKED EXAMPLE

# Nine salaries scraped from Glassdoor

Sample size  **$n = 9$**

Sample mean **\$92,533**

Std. error **\$4,644**

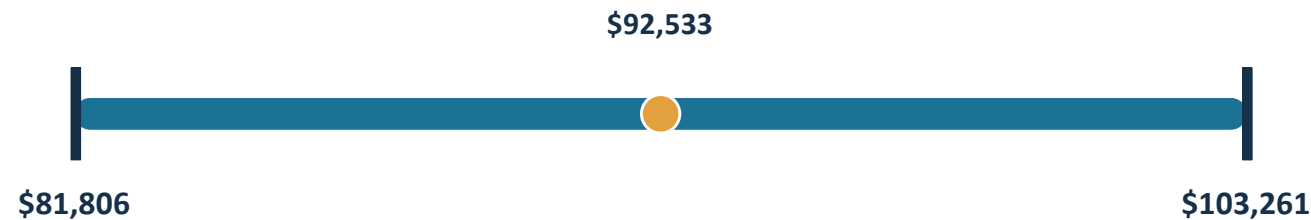
Deg. freedom **8**

- No population variance available, and only nine observations
- Degrees of freedom =  $n - 1 = 8$
- 95% confidence, so  $\alpha = 0.05$  and  $\alpha/2 = 0.025$
- The t-table gives a critical value of 2.31
- Some t-tables carry a CI row, so the 95% column can be read directly instead of computing  $\alpha$

**2.31**

t at 8 df, 95%

$$92,533 \pm 2.31 \times 4,644$$





# The cost of not knowing the variance

Variance known ·  $n = 30$



Variance unknown ·  $n = 9$



## The lesson

- Knowing the population variance produces a narrower interval
- Not knowing it means higher uncertainty, shown as wider boundaries
- Predictions are still possible without it, simply less precise
- The t-statistic is the correct choice here, not z

# Margin of error

The piece of the formula after the  $\pm$  sign has its own name. It determines the whole span of the interval.

$$\text{true population mean} = \bar{x} \pm \text{margin of error}$$

Variance known

$$z_{\alpha/2} \times \sigma / \sqrt{n}$$

Variance unknown

$$t_{n-1, \alpha/2} \times s / \sqrt{n}$$

Smaller margin of error  $\rightarrow$  narrower interval  $\rightarrow$  a better prediction. It is also the one part under the researcher's control.

# What actually moves the margin of error



## Confidence level

Higher confidence raises the statistic, which sits in the numerator. Bigger statistic, wider interval.

*Grades run F to A at 100% confidence. Drop to 99% and it tightens to F+ to A-.*



## Standard deviation

Lower dispersion means the data huddles around the mean, so the estimate has a better chance of landing right.

*"Class mean is B" with grades from A to F, versus "class mean is B and the lowest is C".*



## Sample size

Sample size sits in the denominator, so more observations shrink the margin.

*Three classmates who all scored A's suggest below-average work. All 30 reveal the truth.*

*Only one of the three is truly free: collecting more data.*

# The map of the remaining cases

## **D** Dependent samples

Same subjects measured twice, or subjects linked to each other.  
One formula covers it.

## **I** Independent samples

Values in one sample reveal nothing about the other. Splits into three sub-cases.

### Three sub-cases of independent samples

#### **1** Variance known use z

#### **2** Unknown, assumed equal pooled variance, use t

#### **3** Unknown, assumed different use t, messy degrees of freedom

# Where they show up

1

## Same subject over time

Weight loss, blood tests. One person is measured before and after.

2

## Couples and families

Habits of husbands and wives overlap. Shared time, shared household income.

3

## Same people, different things

Cause and effect rather than before and after. An applicant's SAT score and their admission outcome.

## The trick that makes it easy

Work with the difference per subject instead of two separate variables. Two columns collapse into one, and the data behaves like a single population again — so the familiar one-population formula applies.

$$\bar{d} \pm t_{n-1, \alpha/2} \times s^d / \sqrt{n}$$

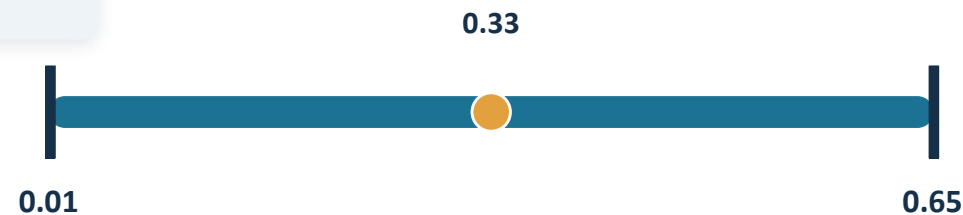
## WORKED EXAMPLE

# Does the magnesium pill work?

A drug company tests a pill meant to raise blood magnesium. Ten people are measured before and after. A healthy reading sits between 1.7 and 2.2 mg/dL.

|                 |               |
|-----------------|---------------|
| Sample size     | <b>n = 10</b> |
| Mean difference | <b>0.33</b>   |
| Std. deviation  | <b>0.45</b>   |

- Populations are normally distributed — with biology this is usually a safe assumption
- Only 10 observations and no population variance, so the statistic is t
- 9 degrees of freedom at 95% confidence gives 2.26



The entire interval is positive, so the true mean difference is positive. With 95% certainty, magnesium levels went up. Based on this small sample, the drug most likely works.

# Case 1 · population variances known

Comparing average grades in two UK university departments. Grades are percentages.

## Engineering

$n = 100$

mean 58%

$\sigma = 10$

## Management

$n = 70$

mean 65%

$\sigma = 5$

- Populations are normally distributed, a fair assumption for grades
- Sample sizes differ, which is completely normal in the real world
- One student's grade cannot affect another department's, so the samples are truly independent
- Big samples plus known variances points to z, not t

$$(\bar{x} - \bar{y}) \pm z_{\alpha/2} \times \sqrt{(\sigma_x^2/n_x + \sigma_y^2/n_y)}$$

The variance of the difference adds the two pieces together, because dispersion is additive — more variables means more variability.

## WORKED RESULT

# Engineering minus management

95% confident the true mean difference between the two departments falls in this interval



### Reading the sign

The whole interval is negative because the difference is engineering minus management, and engineers consistently scored lower.

### Flip the order

Management minus engineering gives 4.72 to 9.28. Perfectly symmetrical around zero. Either direction is valid as long as it stays consistent.



## Case 2 · variances unknown, assumed equal

Are apples cheaper in LA than in New York? Prices are collected from 10 grocery shops in New York and 8 in LA.

### New York

$n = 10$

\$3.94

$s = \$0.18$

### Los Angeles

$n = 8$

\$3.25

$s = \$0.27$

- Population variances are unknown but assumed to be the same in both cities
- They are therefore estimated with the pooled sample variance, an unbiased estimator built only from the two sample sizes and standard deviations
- Pooled variance = 0.05, pooled standard deviation = \$0.22

$$(\bar{x} - \bar{y}) \pm t_{n_1+n_2-2} \times \sqrt{(s^2/n_1 + s^2/n_2)}$$

Earlier cases used  $n - 1$  because there was one sample and one variable. Here there are two of each.

### Degrees of freedom

total sample size – number of variables  
 $10 + 8 - 2 = 16$

WORKED RESULT

## Apples: New York minus LA

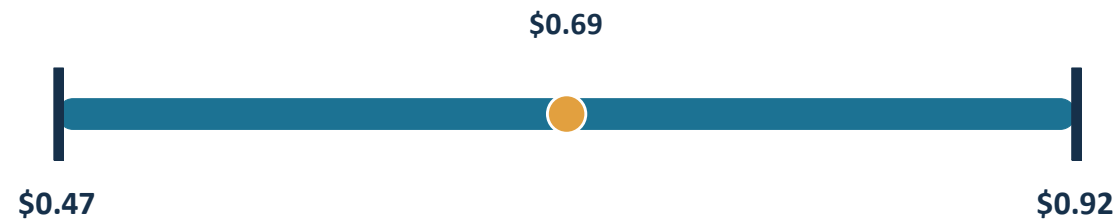
2.12

t at 16 df, 95%

$$(3.94 - 3.25) \pm 2.12 \times SE$$

16

degrees of freedom



At 95% confidence, the true difference in apple prices between New York and LA lies between 47 and 92 cents. Apples in New York are clearly more expensive.

## Case 3 · variances unknown, assumed different

*The case where each population is allowed to keep its own variance.*

$$(\bar{x} - \bar{y}) \pm t \times \sqrt{(s^2_1/n_1 + s^2_2/n_2)}$$

1

### Same shape

Still the difference of two sample means, plus or minus a statistic times a standard error.

2

### No pooling

Each variable keeps its own sample variance, divided by its own sample size.

3

### Ugly df

A formula for the degrees of freedom exists, but it is not worth memorising. Software handles it.

In practice this case appears inside software output rather than on paper.

# Which confidence interval formula?

|                    |                             |                                                    |
|--------------------|-----------------------------|----------------------------------------------------|
| ● One population   | $\sigma$ known              | $z \cdot \sigma / \sqrt{n}$                        |
| ● One population   | $\sigma$ unknown            | $t_{n-1} \cdot s / \sqrt{n}$                       |
| ● Two, dependent   | work with differences       | $t_{n-1} \cdot s^d / \sqrt{n}$                     |
| ● Two, independent | $\sigma$ known              | $z \cdot \sqrt{(\sigma^2_1/n_1 + \sigma^2_2/n_2)}$ |
| ● Two, independent | $\sigma$ unknown, equal     | $t_{n_1+n_2-2} \cdot \text{pooled variance}$       |
| ● Two, independent | $\sigma$ unknown, different | $t \cdot \text{each variance separately}$          |

*Six branches cover every confidence interval in this unit.*

## PRACTICAL EXAMPLE

# Al Bundy Shoe Shop

*Using confidence intervals to solve an inventory problem*

- US company, 30 years old, also operating in Canada, the UK and Germany
- Sells mid to high end shoes priced \$120 to \$200
- The problem: too much unsold inventory collecting dust on the shelves
- This is worse than understocking — the money is already spent on product that will not move
- Data covers 2014 to 2016: invoice, date, country, shop, gender, shoe size, unit price, discount

### First question

Sample or population?

Three years of sales is obviously a sample, not the population of all sales.

# Getting to know the data first

1

## Split by gender

Men's and women's shoes are completely different products. Bundling them yields deceiving predictions — different feet, different types, different models.

2

## Count frequencies by size

The problem is about inventory, so divide the stock by shoe size and count how often each one sells.

3

## Add the country dimension

Three dimensions in total: shoe size, country and gender. Two tables were built, one per gender.

4

## Justify normality

Sales are averaged over a long-running business, so the central limit theorem applies and inference is safe.

*None of this is calculation. Understanding the dataset is the first step.*

# How many pairs should be stocked?

Take the last 12 months of men's sales and build a 95% confidence interval for each shoe size. Because people have different sized feet, that means 17 separate intervals.

12

observations per size

11

degrees of freedom

2.20

t at 95%

17

intervals to build

## The decision rule

**Stock the number closest to the UPPER limit of each interval. Running out then becomes very unlikely, without leaving shoes sitting in storage indefinitely.**

- Four pairs of men's size 6, three pairs of size 6.5, and so on down the list
- Heaviest stocking goes to size 9.5
- Size 16 is predicted to yield no sales at all

# Are these two shops selling the same?

Women's shoe sales in two German shops, GER1 and GER2, using 2016 data.

## Independent?

Yes. The same customers do not shop both branches of the same brand in one year. Exceptions exist but are not the norm.

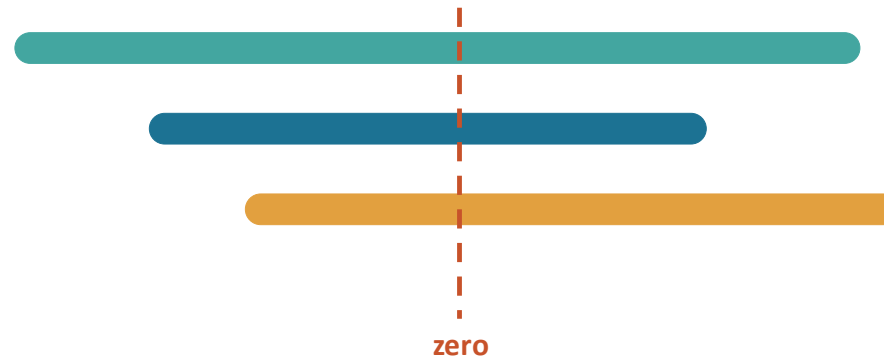
## Variances?

Unknown, but the same market in the same country, so they are assumed equal and pooled.

## Statistic?

t with  $12 + 12 - 2 = 22$  degrees of freedom. At 95% confidence that is 2.07.

15 intervals came out, and zero sits inside all of them



Except for the unsellable size 4, every interval starts negative and finishes positive. Neither shop can be shown to outsell the other for any size.



# “No result” is still a result

*The result looks empty. It is not.*

## 1 Not one-sided

The intervals do not consistently favour either shop. Some lean negative, others lean positive, so it flips by size.

## 2 Practically identical

The two shops are so balanced they could share a warehouse or swap stock to improve results.

## 3 Safe to bundle

They can be treated as one unit for any analysis, action or decision. On average they move together.

## 4 A future alarm

If one starts clearly outperforming later, that is a real signal something new is happening.

## PART 2

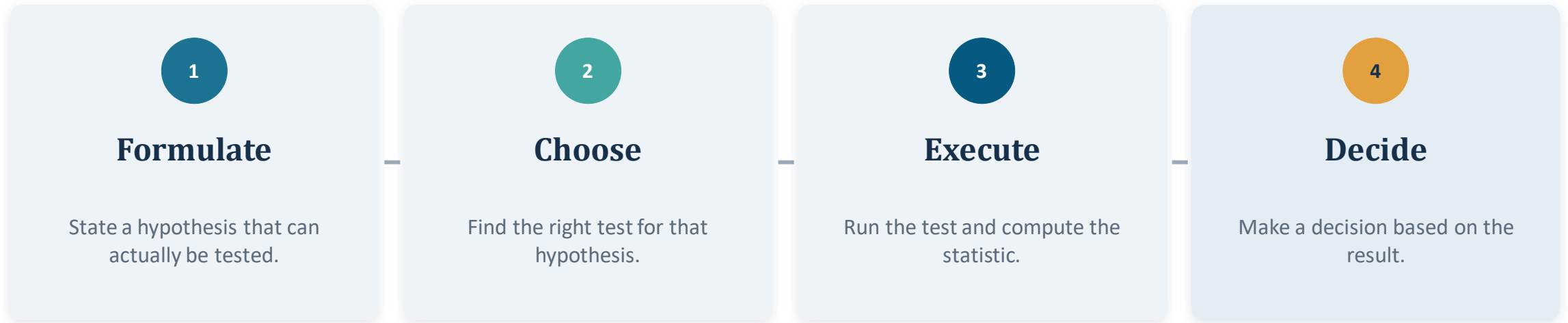
# Hypothesis Testing

*When the situation calls for an answer, not a range.*

- Null and alternative hypotheses
- Significance levels, rejection regions and p-values
- Type I and Type II errors, and the power of a test

# Estimation is not a decision

Confidence intervals indicate where a parameter is likely to be. Acting on it requires a yes or no, which is what a test provides.



*The four steps of data-driven decision making*

# What makes a hypothesis testable

**A hypothesis is an idea that can be tested.**

## Not yet a hypothesis

“Apples in New York are expensive.”

It is a statement, but there is nothing to compare it against.

## Now it is one

“Apples in New York cost more than \$1.75 per pound.”

Define the comparison and it becomes testable.

## The data test

“Would the USA do better under one administration than another?” is an idea, but no data exists for the thing that never happened, so it cannot be a statistical hypothesis. Comparing two completed administrations can be, because both have data.

# The null and the alternative



## Null hypothesis

The statement being tested. The status quo. Example: the mean data scientist salary is \$113,000.



## Alternative hypothesis

Everything else. The challenge to the status quo. Example: the mean salary is not \$113,000.

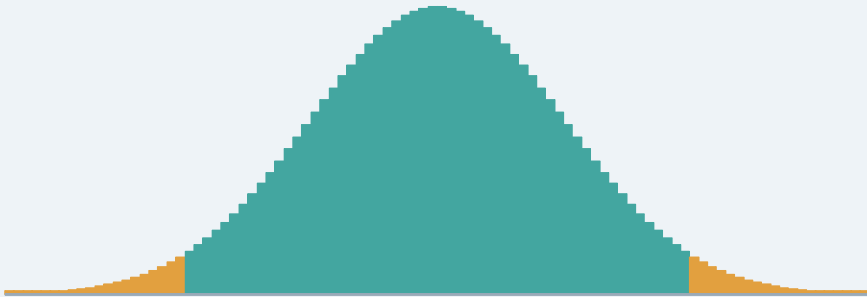
### The mindset that trips everyone up

- Innocent until proven guilty. The null is assumed true and the evidence must prove otherwise
- The researcher is generally trying to REJECT the null hypothesis
- Null = the present state of affairs. Alternative = the proposition, the change, the innovation
- Test outcomes refer to the population parameter, not to the sample statistic

# One-sided or two-sided

## Two-sided

A difference in either direction matters.



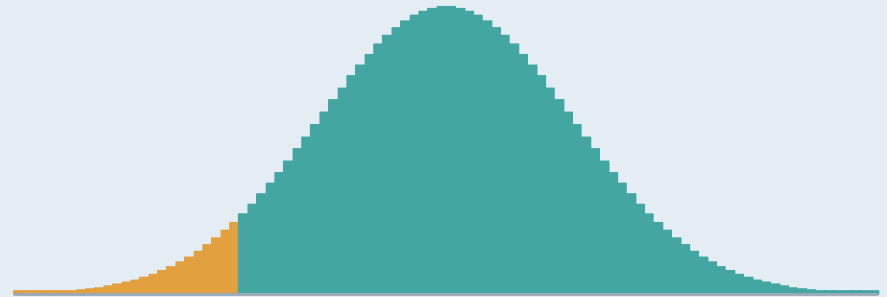
$$H_0: \mu = \$113,000$$

$$H_1: \mu \neq \$113,000$$

*Rejection region is split across both tails.*

## One-sided

Only one direction matters.



$$H_0: \mu \geq \$125,000$$

$$H_1: \mu < \$125,000$$

*The claim under challenge: data scientists earn more than \$125,000.*

## Choosing the significance level $\alpha$

**$\alpha$  = the probability of rejecting the null when it is actually true**

Typical values are 0.01, 0.05 and 0.1, chosen according to how much certainty the situation demands.

### **Be strict: $\alpha = 0.01$**

A machine filling the 12-ounce Coca-Cola bottle. Pour 12.1 ounces and liquid spills and the label is ruined. Almost no mistakes can be tolerated.

### **Be relaxed: $\alpha = 0.05$ or $0.1$**

Predicting how much Coca-Cola people drink on average. Humans and companies behave randomly, so a higher degree of error is expected and 12 versus 12.1 ounces hardly matters.

### STEP 3

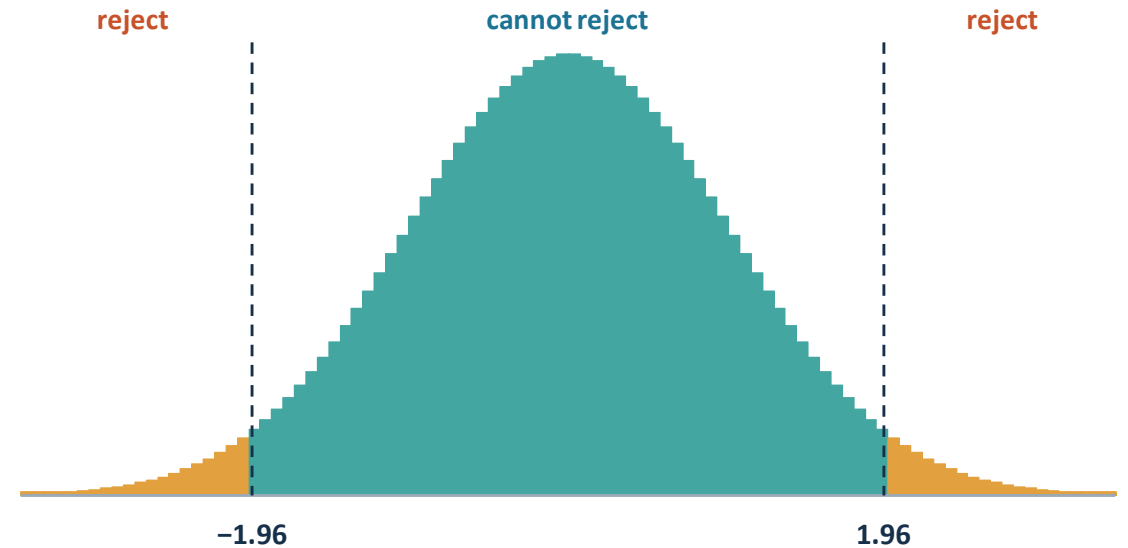
# The rejection region

A university dean believes students average a 70% GPA. The claim is tested.

$$Z = (\bar{x} - \mu_0) / \text{standard error}$$

- The sample mean is standardised against the hypothesised value  $\mu_0$
- Sample mean close to  $\mu_0 \rightarrow Z$  near zero  $\rightarrow$  accept the null
- Sample mean far away  $\rightarrow Z$  far from zero  $\rightarrow$  reject
- If they are exactly equal,  $Z$  is exactly zero

**Decision rule:**  $|Z| > \text{critical value} \rightarrow \text{reject } H_0$

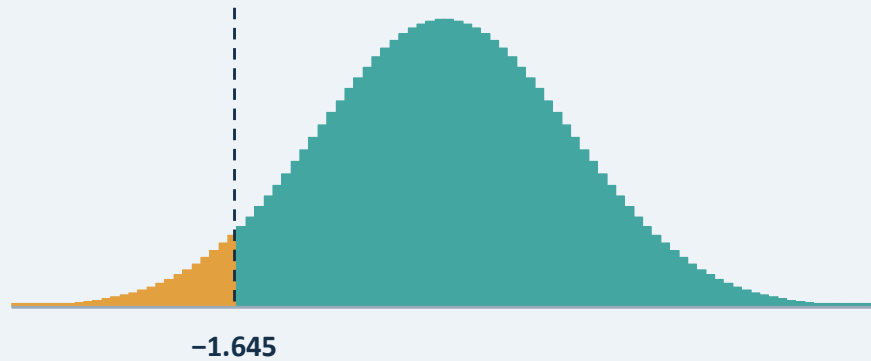


With  $\alpha = 0.05$ , each shaded tail holds 0.025 of the area. Land in a tail and the null is rejected.



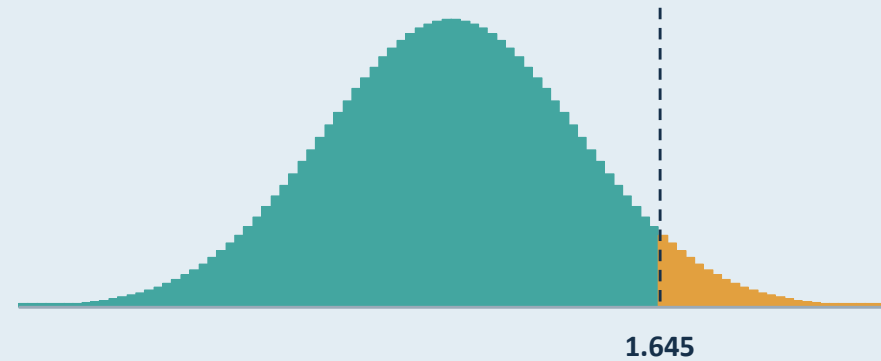
# All of $\alpha$ goes to one side

Left tail



$H_0: \mu \geq \$125,000$ . A Z below  $-1.645$  rejects the null, giving evidence that salaries are under \$125,000.

Right tail



$H_0: \mu \leq 70\%$ . A test statistic above the cutoff rejects the null. Otherwise it stands.

# Type I and Type II errors

|              | $H_0$ is true                                                                                                      | $H_0$ is false                                                                                    |
|--------------|--------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------|
| Accept $H_0$ | <p><b>Correct</b></p> <p>The right call, with no cost attached.</p>                                                | <p><b>Type II error</b></p> <p>A false null was accepted. Probability <math>\beta</math>.</p>     |
| Reject $H_0$ | <p><b>Type I error</b></p> <p>A true null was rejected. A false positive.<br/>Probability <math>\alpha</math>.</p> | <p><b>Correct</b></p> <p>The goal. Probability <math>1 - \beta</math>, the power of the test.</p> |

- $\alpha$  is chosen by the researcher, so the Type I error is entirely their responsibility
- $\beta$  depends mainly on sample size and the size of the effect — hard sampling or a tiny effect makes Type II more likely

# The four outcomes, in everyday terms

A student is unsure whether a classmate returns their interest.  $H_0$ : the classmate is not interested. Accepting the null means doing nothing. Rejecting it means asking them out.

## Accept a true null

The classmate is not interested and nothing happens.  
Embarrassment avoided.

## Reject a false null

The interest is mutual and the invitation follows. The best outcome.

## Type I error

The classmate is not interested but the invitation goes out anyway.  
A false positive, and an awkward conversation.

## Type II error

The interest was real, but the null was accepted and nothing happened. The chance is gone.

*Statistics is very useful, but it is not perfect. Both errors happen.*

## TEST 1

# One mean, variance known

Glassdoor reports a mean data scientist salary of \$113,000, based on self-reported numbers. The claim is checked against a sample.

Sample mean **\$100,200**

Population  $\sigma$  **\$15,000**

Sample size **30**

$H_0: \mu_0 = \$113,000$

$H_1: \mu_0 \neq \$113,000$

Two-sided, because a difference in either direction matters.

**-4.67**

the Z-score

### Two different Zs — do not mix them up

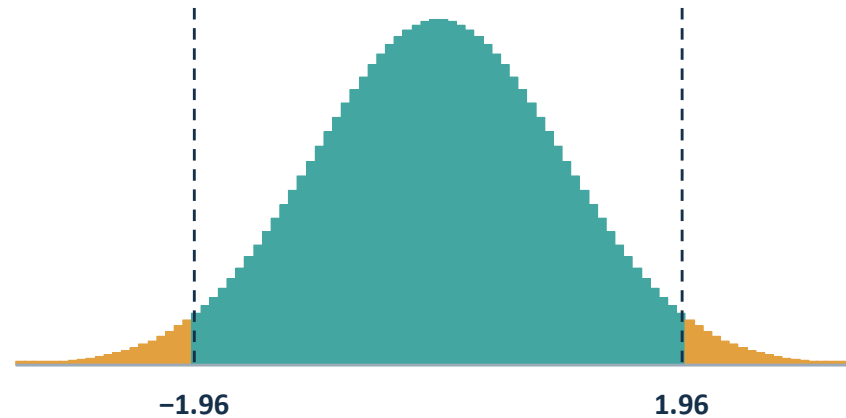
- Z-score: the standardised variable computed from the test
- Critical value: the z read from the table
- Compare absolute values — easier with positives, and some tables have no negatives at all

## TEST 1 RESULT

# Rejected, and comfortably so

1.96

critical value at 5%



2.58

critical value at 1%

*A Z-score of 4.67 lands far outside this picture entirely.*

### At 5% significance

Critical value 1.96.  $4.67 > 1.96$ , so  $H_0$  is rejected. There is no statistical evidence that the mean salary is \$113,000.

### At 1% significance

Critical value 2.58.  $4.67 > 2.58$ , so it is rejected here too. How much further can the level be tightened before rejection fails?

# The smallest $\alpha$ that still permits rejection

**Reject  $H_0$  when p-value < significance level**

- Instead of testing at preassigned levels, find the exact cutoff for the observed statistic
- For the salary test the Z-score was  $-4.67$ . Most tables do not even reach that far, so rounding to the nearest available value gives roughly  $0.0001$
- That result is extremely significant by any measure

The closer to zero, the more significant the result. Researchers hope to see three zeros after the decimal point.

## Reading it off the table

Say the Z-score is  $2.12$ . The table value is  $0.983$ .

**One-sided  $\rightarrow 1 - 0.983 = 0.017$**

**Two-sided  $\rightarrow 0.017 \times 2 = 0.034$**

*That Z would reject at 5% but not at 1%.*

# One rule for every distribution

1

## Universal

Normal, Student's T, binomial, uniform — whatever the test, the p-value rationale holds unchanged.

2

## Reported by software

Statistical packages report it directly. Judging whether the result is significant is then the researcher's call.

3

## No table hunting

No matching of alphas to table rows and columns is needed.

4

## Honest about strength

A single number carries both the decision and the strength of the evidence.

**One caution: the p-value is best obtained from a digital medium rather than a printed table.**

## TEST 2

# One mean, variance unknown

A company's email open rate is 40%. Is a competitor performing better? Only ten of the competitor's campaign open rates are available, so the sample is small.

**H<sub>0</sub>: mean open rate ≤ 40%**

**H<sub>1</sub>: mean open rate > 40%**

*One-sided. The test is whether it is higher, so the null states the opposite.*

|             |        |
|-------------|--------|
| Sample mean | 37.7%  |
| Sample s    | 13.74% |
| Std. error  | 4.34%  |

**-0.53**

T-score

**1.83**

critical t

### The decision

0.53 is lower than 1.83, and when the absolute T-score falls below the table value the null cannot be rejected. At this significance level there is no statistical basis for claiming the competitor's open rate is higher than 40%.



# Checking with the p-value

0.304

p-value

Far above every common significance level, so the same conclusion.  $H_0$  cannot be rejected.

## Quick check at 1%

If the significance level were 0.01, the p-value of 0.304 would still be higher. Still no rejection.

*Tightening the standard can never turn a failure to reject into a rejection.*

## Rule worth writing down

**A test that cannot be rejected at 0.05 significance cannot be rejected at any smaller level either.**

# Two means, dependent samples

The magnesium pill again. The earlier treatment produced an interval. This time the goal is a single definite verdict on whether the drug works.

$H_0: \mu_{\text{before}} \geq \mu_{\text{after}}$

$H_1: \mu_{\text{before}} < \mu_{\text{after}}$

*Levels are expected to rise, so the null states the opposite.*

*Hypotheses are always built so that the null is the statement to be knocked down.*

## Simplifying with D

$\mu_{\text{before}} - \mu_{\text{after}} = 0$  or positive. Replace that whole expression with  $D_0$ , the hypothesised population mean difference, and restate the hypotheses in terms of D.

**-0.33**

mean difference

**0.14**

standard error

**-2.29**

T-score

**0.024**

p-value

*Small sample, assumed normal population, unknown variance — so the t-statistic, as always in that combination.*

# It depends who is asking

## At $\alpha = 0.05$

The p-value of 0.024 is lower, so the null is rejected. At 5%, the pill works.

## At $\alpha = 0.01$

The p-value is higher, so the null stands. At 1%, the evidence is not there.

## What follows from this

- Choosing the significance level is the researcher's call, and it should fit what is at stake
- For a medicine, caution is warranted, because real people will take it
- If it cannot clear 1%, perhaps take it back to the laboratory
- Better still, test again with a larger sample — 100 people would improve precision significantly

# Two means, independent, variances known

The dean claims management students outperform engineering students by exactly 4 percentage points. That claim is now tested.

$$H_0: \mu_e - \mu_m = -4$$

$$H_1: \mu_e - \mu_m \neq -4$$

*Two-sided. The question is not what the difference is, but whether it is exactly four.*

## Choosing the statistic

- Small samples, unknown variances  $\rightarrow t$
- Large samples, known variances  $\rightarrow z$
- Large samples, unknown variances  $\rightarrow$  researcher's call,  $z$  is generally fine

**-2.44**

Z-score

**0.015**

two-sided p-value

**5%**

significance level

**0.015 < 0.05, so the null is rejected. There is enough evidence that the difference is not 4 percentage points.**

# The sign reveals the direction

A two-sided test says the difference is not what was claimed, but not which way. The sign of the test statistic fills that gap.

## – negative statistic

Reversing the standardisation shows the true value is likely LOWER than the hypothesised one.

## + positive statistic

The true value is likely HIGHER than the hypothesised one.

### Applied to the dean's claim

The Z-score was  $-2.44$ , so the true mean difference is likely below  $-4$  — meaning  $-5$ ,  $-6$  and so on. That is additional insight the same test yields at no extra cost.

## TEST 5

# Two means, independent, unknown but equal variances

The apple prices again. Are New York and LA the same?

$$H_0: \mu_Y = \mu_A$$

$$H_1: \mu_Y \neq \mu_A$$

- Pooled variance again: 0.05, because variances are unknown but assumed equal
- Degrees of freedom:  $10 + 8 - 2 = 16$ . Small samples plus unknown variance means t

6.53

test statistic

0.0000001

two-sided p-value

16

degrees of freedom

### Rules of thumb for reading a statistic

- For common tests, reject when the T-score is bigger than 2
- Any Z or T score above 4 is extremely significant, and the chance of a Type I error is virtually zero

## AN HONEST CAVEAT

# A wildly significant result may mean a pointless question

- Means of \$3.94 and \$3.25 with standard deviations around \$0.20 already make the difference obvious
- No test was needed to see that. The hypothesis was poorly designed, not brilliantly confirmed
- A far more interesting question: is the price of apples in New York 20% higher than in LA?

*Statistical significance is not the same thing as an interesting finding.*

# Which test statistic?

● One mean

$\sigma$  known

$$Z = (\bar{x} - \mu_0)/SE$$

● One mean

$\sigma$  unknown

$$t = (\bar{x} - \mu_0)/SE, df = n - 1$$

● Two means, dependent

work with differences

$$t \text{ on } \bar{d}, df = n - 1$$

● Two means, independent

$\sigma$  known, big samples

**Z on the difference**

● Two means, independent

$\sigma$  unknown, equal

$$t, \text{ pooled variance, } df = n_1 + n_2 - 2$$

Then in every case: compare |statistic| to the critical value, or compare the p-value to  $\alpha$ .



## PRACTICAL EXAMPLE

# Spark Fortress Inc.

*Testing whether a company pays men and women differently*

- A large firm with more than 5,000 employees, working from a sample of 174
- Data covers name, age, gender, nationality, ethnicity, tenure, department, position and annual salary
- Split the sample into two sub-samples, one male and one female
- Two samples drawn from the SAME population can still be independent, as long as values in one reveal nothing about the other
- Regression would be the preferred method, but this analysis uses a test on the mean salary instead

### The test

$$H_0: \mu^m - \mu^f = 0$$

$$H_1: \mu^m \neq \mu^f$$

t-test, independent samples,  
variance unknown and assumed  
equal.

# The whole-sample result

98 / 76

female / male

1.34

T-score

172

degrees of freedom

0.182

p-value

**The p-value is far above every common significance level, so the null cannot be rejected. There is not enough statistical evidence of a wage gap at this firm.**

## A useful side note

With 172 degrees of freedom, Student's T has essentially become the normal distribution. Past 50 df the two overlap, so the p-value for a T-score of 1.34 and a Z-score of 1.34 are virtually the same.

# Then somebody looked at the data by hand

- Sorting salaries from largest to smallest shows the highest paid employee is the president and CEO, Caroline Bold, who is female
- That may reflect a genuinely egalitarian culture. It may also mean one very high salary biased the whole result
- The worry: on average women look equally rewarded, when in fact very few of them are

So segment the data by age and run the same test again

## Below 35

T-score

**0.43**

p-value

**0.668**

Virtually no gender wage gap. The p-value is so large there is nothing to find here.

## Above 35

T-score

**2.00**

p-value

**0.048**

Just below 0.05, so at 95% significance the null is rejected. A wage gap does exist for older employees.

# What can and cannot be said

## Direction of the gap

It was a two-sided test, so the sign trick applies. The T-score of 2.00 is positive, meaning the pay difference favours males.

## Limitations

Position and ethnicity were left out, so the full picture inside the firm remains uncertain.

## The conclusion, stated carefully

Overall there is no wage gap at Spark Fortress, and that result is driven by wage equality among younger employees. That is a good indicator: the company is firmly moving towards complete equality.

**Extension exercise: run the same test for racial discrimination in the firm, using the same dataset.**

# Everything on one page

## Confidence interval

point estimate  $\pm$  reliability factor  $\times$  SE

## One mean, $\sigma$ known

$$\bar{x} \pm z_{\alpha/2} \cdot \sigma/\sqrt{n}$$

## One mean, $\sigma$ unknown

$$\bar{x} \pm t_{n-1, \alpha/2} \cdot s/\sqrt{n}$$

## Dependent samples

$$\bar{d} \pm t_{n-1, \alpha/2} \cdot s^d/\sqrt{n}$$

## Independent, $\sigma$ known

$$(\bar{x} - \bar{y}) \pm z_{\alpha/2} \cdot \sqrt{(\sigma^2_1/n_1 + \sigma^2_2/n_2)}$$

## Independent, equal var.

$$(\bar{x} - \bar{y}) \pm t \cdot \sqrt{(s^2_p/n_1 + s^2_p/n_2)}, \text{ df} = n_1 + n_2 - 2$$

## Test statistic

(sample stat – hypothesised value)/SE

## Decision, critical value

reject  $H_0$  when |statistic| > critical value

## Decision, p-value

reject  $H_0$  when p-value <  $\alpha$

## Power of a test

$1 - \beta$ , raised mainly by a bigger sample

*SE = standard error. Degrees of freedom = total sample size – number of variables.*

# Seven things to carry forward

- 1 A point estimate is a guess. An interval is an honest answer.
- 2 Confidence and precision pull against each other. Both cannot be maximised at once.
- 3 Variance known goes with  $z$  and  $\sigma$ . Variance unknown goes with  $t$  and  $s$ .
- 4 The one lever fully under the researcher's control is sample size.
- 5 The null is built as the statement to be knocked down.
- 6 Reject when the p-value drops below  $\alpha$ . That rule works for every distribution.
- 7 A significant result is not automatically an interesting one.